WORKED SOLUTION

STEP II 2022, Question 2

Pure • Recurrence Relations

STEP II 2022, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Multiply the relation through by 2 and it becomes a statement about *second differences*: $u_{n+2} - 2u_{n+1} + u_n$. In (i) this is zero, so the first differences are constant and the sequence is linear. In (ii) and (iii) the second difference is a given constant or a given linear function of n, so the trick is to subtract off a polynomial with exactly that second difference, leaving a sequence of the type in (i).

Solution

Part (i)

Doubling the relation and rearranging gives, for all $n \geq 1$,

$$u_{n+2} - u_{n+1} = u_{n+1} - u_n.$$

So each difference equals the one before it, and by induction every difference equals the first one, $d = u_2 - u_1$. That is the constant difference between successive terms.

Adding up $n - 1$ of these differences,

$$u_n = u_1 + (n-1)d = dn + (u_1 - d),$$

which is a polynomial in n of degree 1 if $d \neq 0$ and of degree 0 (a constant) if $d = 0$. Either way the degree is at most 1.

Part (ii)

Put $v_n = t_n + pn^2$. Since

$$\tfrac{1}{2}(p(n+2)^2 + pn^2) = p(n^2 + 2n + 2) = p(n+1)^2 + p,$$

the relation for v_n becomes

$$t_{n+1} + p(n+1)^2 = \tfrac{1}{2}(t_{n+2} + t_n) + p(n+1)^2 + p - p,$$

that is, $t_{n+1} = \frac{1}{2}(t_{n+2} + t_n)$ for all $n \geq 1$. By part (i), $t_n = A + Bn$ for some constants A and B, so

$$v_n = pn^2 + Bn + A.$$

Because $p \neq 0$ the n^2 term survives, and v_n has degree exactly 2.

With $v_1 = v_2 = 0$:

$$p + B + A = 0, \qquad 4p + 2B + A = 0.$$

Subtracting, $3p + B = 0$, so $B = -3p$ and then $A = 2p$. Hence

$$v_n = p(n^2 - 3n + 2) = p(n-1)(n-2).$$

Part (iii)

This time no substitution is suggested, so we have to find one. Doubling the relation,

$$w_{n+2} - 2w_{n+1} + w_n = 2(an + b).$$

Ask yourself Which polynomial in n has a second difference that is linear in n?

Each difference lowers the degree of a polynomial by one, so we need a cubic. Write $\Delta^2 f(n) = f(n+2) - 2f(n+1) + f(n)$. Direct expansion gives

$$\Delta^2 n^3 = (n^3 + 6n^2 + 12n + 8) - 2(n^3 + 3n^2 + 3n + 1) + n^3 = 6n + 6,$$
$$\Delta^2 n^2 = (n^2 + 4n + 4) - 2(n^2 + 2n + 1) + n^2 = 2.$$

So $\Delta^2(cn^3 + en^2) = 6cn + 6c + 2e$, and this equals $2an + 2b$ for every n when

$$c = \frac{a}{3}, \qquad e = b - 3c = b - a.$$

Now put $w_n = t_n + \frac{a}{3}n^3 + (b-a)n^2$. Since Δ^2 is linear, $\Delta^2 w_n = \Delta^2 t_n + 2(an + b)$, and the relation becomes $\Delta^2 t_n = 0$, which is exactly the relation of part (i). So $t_n = A + Bn$ and

$$w_n = \frac{a}{3}n^3 + (b-a)n^2 + Bn + A.$$

Since $a \neq 0$, the cubic term is present and w_n has degree 3.

With $w_1 = w_2 = 0$:

$$\tfrac{a}{3} + (b-a) + B + A = 0,$$
$$\tfrac{8a}{3} + 4(b-a) + 2B + A = 0.$$

Subtracting the first from the second gives $\frac{7a}{3} + 3(b-a) + B = 0$, so $B = \frac{2a}{3} - 3b$, and then the first equation gives $A = 2b$. Therefore

$$w_n = \frac{a}{3}n^3 + (b-a)n^2 + \left(\frac{2a}{3} - 3b\right)n + 2b.$$

Since $w_1 = w_2 = 0$, the factors $n-1$ and $n-2$ must appear, and matching the n^3 and constant terms gives the tidier form

$$w_n = \frac{(n-1)(n-2)(an+3b)}{3}.$$

(Expanding $(n-1)(n-2)(an+3b)$ gives $an^3 + (3b-3a)n^2 + (2a-9b)n + 6b$, which agrees.)

Checks

- Setting $a = 0$, $b = p$ in (iii) should recover (ii), and it does: $(n-1)(n-2)\cdot 3p/3 = p(n-1)(n-2)$.
- Run the recurrence directly. With $a = 3$, $b = 0$ we have $w_{n+2} = 2w_{n+1} - w_n + 6n$, so from $w_1 = w_2 = 0$ we get $w_3 = 6$, $w_4 = 24$, $w_5 = 60$. The formula gives $(n-1)(n-2)n$, which is $6, 24, 60$.
- A degree-3 sequence has constant third differences. For $0, 0, 6, 24, 60$ the differences are $0, 6, 18, 36$, then $6, 12, 18$, then $6, 6$: constant, equal to $2a$.

Common slips

- Saying the sequence in (i) has degree exactly 1: if $u_1 = u_2$ it is constant.
- Trying only $w_n = t_n + kn^3$ in (iii). The n^3 term on its own produces $6kn + 6k$, so you also need an n^2 term to fix the constant.
- Forgetting to justify "degree exactly 2" (or 3) from $p \neq 0$ (or $a \neq 0$).

Beyond STEP

This is the discrete analogue of solving $y'' = f(x)$: the general solution is a particular solution plus the solution of the homogeneous equation, and here the homogeneous solutions are the linear sequences $A + Bn$. The same "complementary function plus particular solution" structure governs linear differential equations, and the method of guessing a polynomial particular solution carries over almost word for word. Boyce and DiPrima, *Elementary Differential Equations*, develops the continuous version.

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